

Assignment 34-Python Basics

Question 1: What is Python and why is it widely used in Data Analytics? Mention any three reasons

Answer:

Python:

Python is a high-level, easy-to-learn programming language used for data analysis, automation, web development, and more.

Why it is widely used in Data Analytics:

Easy to Learn & Read

- Simple syntax (like English)
- Beginners can quickly start working with data

Powerful Libraries

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- Libraries like pandas, numpy, matplotlib make data analysis easy and fast

Large Community Support

- Millions of developers
- Easy to find solutions, tutorials, and support online

Question 2: Explain the difference between List and Tuple in Python.

Answer:

Feature	List	Tuple
Mutability	Mutable	Immutable
Syntax	[]	()
Performance	slower	Faster

Use case of list: when data needs to change

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Use case of tuple: when data should remain fixed

Example of both:

List

```
lst = [1,2,3]
```

```
lst[0] = 10 # allowed
```

Tuple

```
tup = (1,2,3)
```

```
tup[0] = 10 # error
```

Question 3: What is a function in Python?

Why are functions useful?

Answer:

Function: A **function** is a block of code that performs a specific task and can be reused multiple times

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Example:

```
def add(a, b):
```

```
    return a + b
```

```
add(2,4)
```

Why functions are useful:

Code Reusability

- Write once, use many times

Better Organization

- Makes code clean and structured

Reduces Repetition

- Avoid writing the same code again and again

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